

ICRA Updates

Prajjwal Dutta

Arizona State University

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Problem Statement

Advection–Diffusion motion model

$$dx_i(t) = (f_{\text{flow}}(x_i(t), t) + v_i(t)) dt + \sigma_{\text{diff}} dW_i(t), \quad (1)$$

$$\dot{v}_i(t) = u_i(t) - \gamma v_i(t), \quad (2)$$

Control-affine analysis model

We use the reduced model

$$\dot{x}_i(t) = f_{\text{flow}}(x_i(t), t) + u_i(t) + d_i(t), \quad \|d_i(t)\| \leq \bar{d}, \quad (3)$$

where $d_i(t)$ captures unmodeled drag, turbulence, and stochastic effects.



Objective

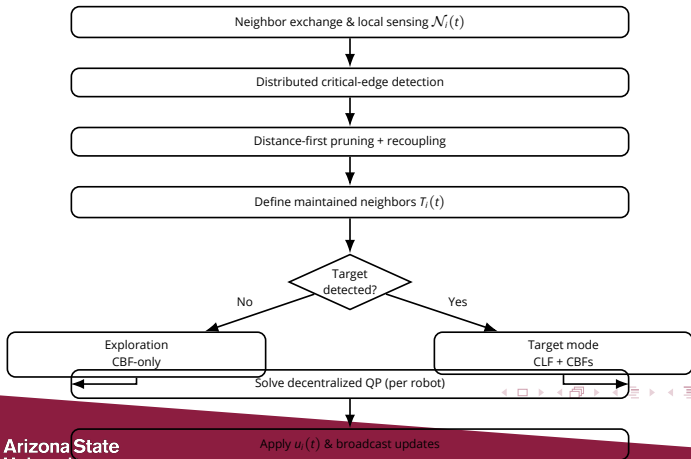
Design distributed controls $u_i(t)$ and a sparse time-varying topology $\mathcal{E}_{\text{topo}}(t) \subseteq \mathcal{E}_t$ such that robots reach targets while guaranteeing, $\forall t \geq 0$,

$$x(t) \in \underbrace{\mathcal{C}_{\text{safe}}}_{\|x_{ij}(t)\| \geq d_{\min}} \cap \underbrace{\mathcal{C}_{\text{conn}}(\mathcal{E}_{\text{topo}}(t))}_{\|x_{ij}(t)\| \leq R_{\max} \quad \forall (i,j) \in \mathcal{E}_{\text{topo}}(t)} . \quad (4)$$

Assumptions

- A1 Bounded control & dominance:** $\|u_i(t)\| \leq u_{\max}$ and $u_{\max} > \bar{d} + L_f R_{\max}, \forall i, \forall t \geq 0$.
- A2 Separation buffer:** $d_{\min} < R_{\max} - \varepsilon$ for some $\varepsilon > 0$.
- A3 Disk-graph communication:** $(i, j) \in \mathcal{E}_t \Leftrightarrow \|x_i(t) - x_j(t)\| \leq R_{\max}$; neighbor-only exchange via $\mathcal{N}_i(t)$; $\mathcal{G}_0$ connected.
- A4 Slow topology variation (consensus/pruning):** $\mathcal{G}_{\text{topo}}(t)$ remains connected and changes slowly enough for required iterations to converge.
- A5 Lipschitz flow field:** $\|f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)\| \leq L_f \|x_i - x_j\|, \forall x_i, x_j, \forall t \geq 0$.

Methodology



Thm 1: Invariance (Proof Sketch I) — Setup & Safety

Relative dynamics and bounds

For $z_{ij} = x_i - x_j$, $r_{ij} = \|z_{ij}\|$,

$$\dot{z}_{ij} = \Delta f_{ij}(t) + (u_i - u_j) + \Delta d_{ij}(t), \quad \|\Delta f_{ij}\| \leq L_f r_{ij}, \quad \|\Delta d_{ij}\| \leq 2\bar{d}.$$

Safety barrier

Let $h_{ij}^{\text{safe}} = r_{ij}^2 - d_{\min}^2$. Then

$$\dot{h}_{ij}^{\text{safe}} = 2z_{ij}^\top \dot{z}_{ij} \geq 2z_{ij}^\top (u_i - u_j) - 2L_f r_{ij}^2 - 4\bar{d} r_{ij}.$$

Enforce the robust CBF condition so that

$$\dot{h}_{ij}^{\text{safe}} \geq -\alpha_s(h_{ij}^{\text{safe}}) \Rightarrow h_{ij}^{\text{safe}}(t) \geq 0 \quad \forall t \geq 0.$$



Thm 1: Invariance (Proof Sketch II) — Connectivity & Intersection

Connectivity barrier

Let $h_{ij}^{\text{conn}} = R_{\max}^2 - r_{ij}^2$. Then

$$\dot{h}_{ij}^{\text{conn}} = -2z_{ij}^T \dot{z}_{ij} \geq -2z_{ij}^T (u_i - u_j) - 2L_f r_{ij}^2 - 4\bar{d} r_{ij}.$$

Enforce the robust CBF condition so that

$$\dot{h}_{ij}^{\text{conn}} \geq -\alpha_c(h_{ij}^{\text{conn}}) \Rightarrow h_{ij}^{\text{conn}}(t) \geq 0 \quad \forall t \geq 0.$$

Conclusion

Each enforced constraint set $\{h_{ij}^{\text{safe}} \geq 0\}$ and $\{h_{ij}^{\text{conn}} \geq 0\}$ is forward invariant; therefore their intersection (over $(i, j) \in \mathcal{E}_s$ and $(i, j) \in \mathcal{E}_c$) is forward invariant.

Thm 2: Distributed Consensus & Connectivity-Preserving Sparsification

Setting

Disk graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$, $(i, j) \in \mathcal{E}(t) \Leftrightarrow \|x_i - x_j\| \leq R_{\max}$. Assume $\mathcal{G}(t_0)$ connected and Assumptions A3–A4 (neighbor-only, quasi-static during consensus/pruning).

Claim (3 parts)

1. **Consensus:** adjacency estimates $\mathbf{A}^l(k)$ updated by (??) satisfy $\|\mathbf{A}^l - \mathbf{A}^p\|_F < 2\epsilon_{\text{conv}} \Rightarrow$ all robots extract the same edge set $\mathcal{E}_{\text{cons}}$ via threshold $\theta_{\text{edge}} \gg 2\epsilon_{\text{conv}}$.
2. **Structure:** running the same deterministic routine on $(\mathcal{V}, \mathcal{E}_{\text{cons}})$ yields identical connected spanning subgraph $\mathcal{G}_C = (\mathcal{V}, \mathcal{C})$.
3. **Pruning:** distance-first pruning on $\mathcal{E} \setminus \mathcal{C}$ with guards (alt-path, degree, margin, backbone, batch-feasibility) preserves connectivity each round and terminates with sparse $\mathcal{G}' = (\mathcal{V}, \mathcal{E}')$, $\mathcal{C} \subseteq \mathcal{E}'$, $|\mathcal{E}'| \geq N - 1$.

Thm 2 Proof Sketch (I–II): Consensus $\rightarrow$ Shared Graph & Backbone

Part I: Edge-set agreement via consensus

Consensus update is a standard Laplacian-driven averaging with correction. If $\lambda_2(\mathbf{L}_{\text{comm}}(t)) > 0$ during the window and topology is quasi-static (A4), then $\mathbf{A}^l(k)$ converge to a common value:

$$\|\mathbf{A}^l(k) - \mathbf{A}^p(k)\|_F < 2\epsilon_{\text{conv}}, \quad \forall l, p.$$

Thresholding with $\theta_{\text{edge}} \gg 2\epsilon_{\text{conv}}$ eliminates ambiguity, hence $\mathcal{E}_{\text{cons}}^l = \mathcal{E}_{\text{cons}}^p = \mathcal{E}_{\text{cons}}$ for all robots.

Part II: Graph structure agreement and spanning connectivity

All robots run the same deterministic backbone extraction on the identical input graph $(\mathcal{V}, \mathcal{E}_{\text{cons}})$, so they output the same set $\mathcal{C}$. By construction $\mathcal{G}_{\mathcal{C}} = (\mathcal{V}, \mathcal{C})$ is a connected spanning subgraph. The backbone guard ensures $\mathcal{C}$ is never removed.

Thm 2 Proof Sketch (III): Safe Distributed Pruning Preserves Connectivity

Induction over pruning rounds

Let $G_k = (\mathcal{V}, \mathcal{E}_k)$ be the graph before round k , with G_0 connected. Assume G_k connected. Consider any removed edge $e = (i, j) \in \mathcal{E}_k \setminus \mathcal{C}$:

- **Alt-path guard:** there exists $P_{\text{alt}} : i \rightarrow \dots \rightarrow j$ in $G_k \setminus \{e\}$ (and avoiding the batch if used), so removing e does not disconnect i and j .
- **Degree guard:** $\deg(i), \deg(j) \geq 2$ pre-removal $\Rightarrow$ no isolated vertex after removal.
- **Backbone guard:** edges in $\mathcal{C}$ (a connected spanning subgraph) are never removed $\Rightarrow$ global connectivity retained.
- **Next-step margin:** $\ell \leq R_{\text{max}} - \varepsilon_{\text{margin}}$ keeps certifying paths valid during quasi-static window (A4).

Thus G_{k+1} remains connected; by induction all rounds preserve connectivity.

Termination + sparsity

Each round removes at least one edge from $\mathcal{E} \setminus \mathcal{C}$ and never removes $\mathcal{C}$. Since any connected graph needs $\geq N - 1$ edges and $\mathcal{C}$ spans $\mathcal{V}$, pruning terminates in finite K with $\mathcal{C} \subseteq \mathcal{E}'$ and $|\mathcal{E}'| \geq N - 1$.



Thm 3: CLF–CBF Practical Goal Convergence (Statement)

CLF–CBF QP

$V_i = \|x_i - x_i^{\text{goal}}\|^2$. Solve

$$\min_{u_i, \gamma_i} \|u_i\|^2 + \lambda \gamma_i^2$$

$$\text{s.t. } \dot{V}_i + \alpha V_i \leq \gamma_i, \quad \|u_i\| \leq u_{\max}, \quad \dot{h}_{ij}^{\text{safe}} + \alpha_s h_{ij}^{\text{safe}} \geq 0 \quad \forall j \in \mathcal{N}_i, \quad \dot{h}_{ij}^{\text{conn}} + \alpha_c h_{ij}^{\text{conn}} \geq 0 \quad \forall j \in \mathcal{C}_i.$$

Claims

Feasible $\forall t$; preserves safety & critical-edge connectivity; and

$$\limsup_{t \rightarrow \infty} \|x_i - x_i^{\text{goal}}\| \leq \epsilon_{\text{goal}} = o\left(\frac{\|f_{\text{flow}}\|_{\max} + \bar{d}}{\sqrt{\alpha}}\right), \quad V_i(t) \leq V_i(0)e^{-\alpha t} \text{ outside residual set.}$$

Thm 3 Proof Sketch (1/3): CBF Feasibility & Constraint Preservation

CBF feasibility

From Thm ?? and Assumption A1 ($u_{\max} > \bar{d} + L_f R_{\max}$), the CBF constraints admit a nonempty feasible set:

$$\mathcal{F}_{\text{CBF}} \neq \emptyset, \quad \mathcal{F}_{\text{CBF}} \subseteq \{u_i : \|u_i\| \leq u_{\max}\}.$$

Immediate consequence

If the QP enforces the CBF inequalities for all t , then

$$\|x_i(t) - x_j(t)\| \geq d_{\min} \quad \forall (i, j) \in \mathcal{E}_s, \quad \|x_i(t) - x_j(t)\| \leq R_{\max} \quad \forall (i, j) \in \mathcal{E}_c.$$

Thm 3 Proof Sketch (2/3): QP Feasibility & Exponential Decay

QP feasibility via CLF relaxation

$$\dot{V}_i = 2(x_i - x_i^{\text{goal}})^\top (f_{\text{flow}}(x_i, t) + d_i + u_i).$$

For any $u_i \in \mathcal{F}_{\text{CBF}}$,

$$|\dot{V}_i| \leq 2\sqrt{V_i}(\|f_{\text{flow}}\|_{\max} + \bar{d} + u_{\max}) \Rightarrow \exists \gamma_i < \infty \text{ s.t. } \dot{V}_i + \alpha V_i \leq \gamma_i.$$

Hence the CLF constraint is always satisfiable and the QP is feasible.

When CBF constraints are inactive

If $\gamma_i^* = 0$ (CLF satisfied without relaxation), then

$$\dot{V}_i \leq -\alpha V_i \Rightarrow V_i(t) \leq V_i(0)e^{-\alpha t}.$$



Thm 3 Proof Sketch (3/3): Residual Set & Practical Convergence

Residual set radius

Near the goal, worst-case

$$\dot{V}_i \lesssim 2\sqrt{\tilde{V}_i}(\|f_{\text{flow}}\|_{\max} + \bar{d}), \quad \bar{\gamma} \leq \frac{4(\|f_{\text{flow}}\|_{\max} + \bar{d})^2}{\alpha}.$$

Hence

$$\epsilon_{\text{goal}} = \sqrt{\frac{\bar{\gamma}}{\alpha}} = \frac{2(\|f_{\text{flow}}\|_{\max} + \bar{d})}{\sqrt{\alpha}}.$$

Shifted Lyapunov argument

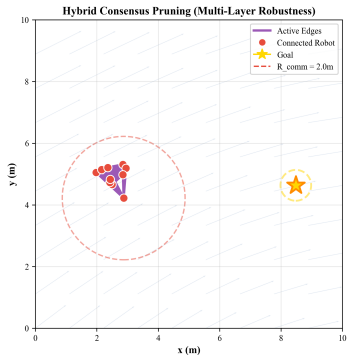
Let $\tilde{V}_i = V_i - \bar{\gamma}/\alpha$. Since $\dot{V}_i + \alpha V_i \leq \gamma_i \leq \bar{\gamma}$,

$$\dot{\tilde{V}}_i \leq -\alpha \tilde{V}_i.$$

Navigation icons: back, forward, search, etc.

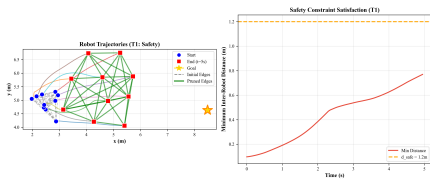
Therefore trajectories converge to $\{x_i : V_i(x_i) \leq \bar{\gamma}/\alpha\}$, i.e., $\limsup_{t \rightarrow \infty} \|x_i(t) - x_i^{\text{goal}}\| \leq \epsilon_{\text{goal}}$.

Sim Setup & Flow Validation

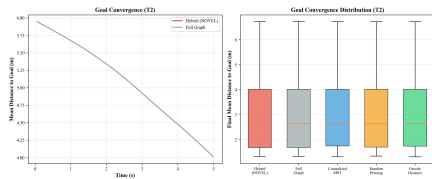


Composite flow field is spatially varying; Lipschitz bound (L_f) validated; initial dense graph shown.

Safety/Connectivity + Goal Convergence



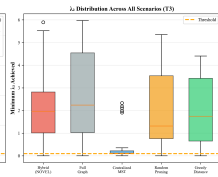
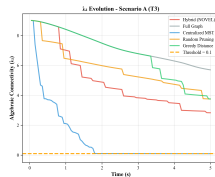
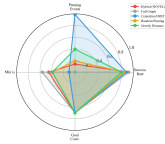
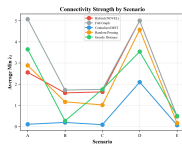
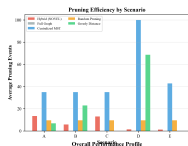
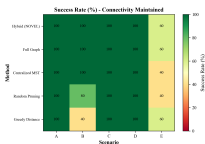
Zero safety violations; critical-edge distances stay $< R_{\text{max}}$.



Goal distance decreases; γ_i activates only when CBFs bind.

Connectivity + Baseline Comparison

Comprehensive Method Comparison Across All Scenarios



Hybrid keeps higher λ_2 than MST while using fewer edges than Full.

Hybrid success and edge reduction across scenarios.

125 Simulation & Performance

Table 1: Scenario parameters.

Sc.	N	$R_{\max}$	Δt
A	10	3.0	0.05
B	10	3.5	0.10
C	10	2.0	0.03
D	20	3.0	0.05
E	12	2.2	0.08

A: nominal, B: large Δt , C: tight R, D: scalability, E: stress.

Table 2: Performance (125 runs).

Method	Succ.	λ_2	Edge red.
Hybrid	92.0	2.45 ± 1.52	29.5
Full	92.0	3.05 ± 1.88	0.0
MST	88.0	0.58 ± 0.83	72.1
Random	84.0	2.34 ± 1.54	33.5
Greedy	80.0	2.43 ± 1.26	28.1

Hybrid matches Full success with $\sim 30\%$ fewer edges; MST is sparse but fragile (λ_2 low).